

Instructor
Prof. B. Stellato

Alis
I. Wang, V. Ranjan, S. Clarke, J. Ge

ORF307: Optimization

Spring 2023 2nd Midterm Exam Solutions

April 18, 11:00am – 12:20pm

Exam Rules

1. Please write out and sign the following pledge on the solution booklet: “I pledge my honor that I have not violated the Honor Code or the rules specified by the instructor during this exam.”
2. You are allowed a single sheet of paper, double-sided, hand-written or typed.
3. You cannot communicate with anyone during the exam.
4. All questions should be answered in the solution booklet. Write your name on every page that you use to write the solutions. Start each problem on a separate page. Answering multiple subparts of the same problem on the same page is fine. We will **not** grade anything written on the exam questions; we will only grade what you write in the solution booklet.
5. You should **always explain your answers** in the solution booklet.
6. There are 3 problems worth a total of 100 points.
7. At the end of the exam, please turn in both your solution booklet and the exam questions.

Name: _____

1. *Problem 1 (35 points)*. An ℓ_∞ constrained problem.

Consider the following optimization problem:

$$\begin{aligned} & \text{minimize} && c^T x \\ & \text{subject to} && \|Ax - b\|_\infty \leq 1, \end{aligned} \tag{1}$$

for $c \in \mathbf{R}^n$, $A \in \mathbf{R}^{m \times n}$, $b \in \mathbf{R}^m$.

- (a) (10 points) Write problem (1) as an LP.
 (b) (15 points) Take the dual of the LP you derived in part (a). Please simplify your final answer to be in terms of c, A, b , and any new dual variables you create.

Now, let us plug in the following values to problem (1):

$$c = (-1), \quad A = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad b = (3, 4).$$

For these values, the dual problem is equivalent to

$$\begin{aligned} & \text{maximize} && -d^T y \\ & \text{subject to} && Gy = h \\ & && y \geq 0, \end{aligned} \tag{2}$$

where

$$d = (3, 4, -3, -4, 1), \quad G = \begin{bmatrix} 1 & 2 & -1 & -2 & 0 \\ -1 & -1 & -1 & -1 & 1 \end{bmatrix}, \quad h = (1, 0).$$

For these values, the optimal solution to (2) is given by $y^* = (0, 0.5, 0, 0, 0.5)$.

- (c) (10 points) For the given problem data, what is the optimal basis B ? What is the optimal basis matrix G_B ? What is y_B^* ?

Solution.

- (a) We introduce an auxiliary variable $t \in \mathbf{R}$. The equivalent LP can be written as:

$$\begin{aligned} & \text{minimize} && c^T x, \\ & \text{subject to} && Ax - b \leq t\mathbf{1}, \\ & && -(Ax - b) \leq t\mathbf{1}, \\ & && t \leq 1, \end{aligned} \tag{3}$$

where $\mathbf{1} \in \mathbf{R}^m$ is the all-ones vector.

Note: It was actually not necessary to include the auxiliary variable t . Therefore, the following is also correct:

$$\begin{aligned} & \text{minimize} && c^T x, \\ & \text{subject to} && Ax - b \leq \mathbf{1}, \\ & && -(Ax - b) \leq \mathbf{1}. \end{aligned} \tag{4}$$

(b) The LP in (3) is correct as is, but to make taking the dual easier, we rewrite it to fit the inequality form:

$$\begin{aligned} & \text{minimize} && c^T x, \\ & \text{subject to} && Ax - t\mathbf{1} \leq b, \\ & && -Ax - t\mathbf{1} \leq -b, \\ & && t \leq 1. \end{aligned} \tag{5}$$

We introduce dual variables $y \in \mathbf{R}^m$, $z \in \mathbf{R}^m$, and $w \in \mathbf{R}$ for each of the respective constraints in (5). The dual becomes:

$$\begin{aligned} & \text{maximize} && -b^T y + b^T z - w \\ & \text{subject to} && A^T y - A^T z = -c, \\ & && -\mathbf{1}^T y - \mathbf{1}^T z + w = 0, \\ & && y, z, w \geq 0. \end{aligned} \tag{6}$$

Note: If you took the approach corresponding to (4), then the dual becomes:

$$\begin{aligned} & \text{maximize} && -(\mathbf{1} + b)^T y - (\mathbf{1} - b)^T z \\ & \text{subject to} && A^T y - A^T z = -c, \\ & && y, z \geq 0. \end{aligned} \tag{7}$$

(c) $B = \{2, 5\}$. $G_B = \begin{bmatrix} 2 & 0 \\ -1 & 1 \end{bmatrix}$. $y_B^* = \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}$.

2. *Problem 2 (35 points)*. Consider the following optimization problem:

$$\begin{aligned}
 &\text{minimize} && -x_1 - 2x_2 \\
 &\text{subject to} && 2x_1 + 3x_2 \leq 3 \\
 &&& -2x_1 + 3x_2 \leq 6 \\
 &&& x_1 \geq 0 \\
 &&& x_2 \geq 0.
 \end{aligned} \tag{1}$$

(a) (5 points) Write problem (1) as an LP of the following form:

$$\begin{aligned}
 &\text{minimize} && c^T x \\
 &\text{subject to} && Ax \leq b.
 \end{aligned} \tag{2}$$

(b) (10 points) Write down the dual problem of the LP you derived in part (a). Expand your answer with the problem data instead of just writing the general dual formulation.

(c) (15 points) Show that feasible $x = (0, 1)$ is optimal.

Hint: Use complementary slackness.

(d) (5 points) Let us modify b in (2) by adding $u = (1, 0, 0, 0)$ to it, i.e., we replace b with $b + u$. Assuming that the optimal basis remains unchanged, what would be the new optimal value, denoted by $p^*(u)$, of (2) with the modified b ?

Solution.

(a) We have

$$c = \begin{bmatrix} -1 \\ -2 \end{bmatrix}, \quad A = \begin{bmatrix} 2 & 3 \\ -2 & 3 \\ -1 & 0 \\ 0 & -1 \end{bmatrix}, \quad b = \begin{bmatrix} 3 \\ 6 \\ 0 \\ 0 \end{bmatrix}.$$

(b) The dual problem is given by

$$\begin{aligned}
 &\text{maximize} && -3y_1 - 6y_2, \\
 &\text{subject to} && 2y_1 - 2y_2 - y_3 - 1 = 0, \\
 &&& 3y_1 + 3y_2 - y_4 - 2 = 0, \\
 &&& y \geq 0.
 \end{aligned} \tag{3}$$

(c) The first and third constraints are active at x . Thus, we consider $y = (y_1, 0, y_3, 0)$. By constraints in (3), we have

$$\begin{bmatrix} 2 & -1 \\ 3 & 0 \end{bmatrix} \begin{bmatrix} y_1 \\ y_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix},$$

which implies $y_1 = 2/3$ and $y_3 = 1/3$. Since all the elements of $y = (2/3, 0, 1/3, 0)$ are non-negative, we obtain a feasible solution for the dual problem (3). Also the duality gap is 0. Thus, we conclude that x is primal optimal.

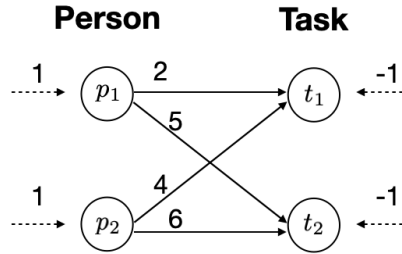
(d) Since the optimal basis remains unchanged, we have

$$\begin{aligned}
 p^*(u) &= -(b+u)^T y^* \\
 &= - \begin{bmatrix} 4 & 6 & 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{2}{3} \\ 0 \\ \frac{1}{3} \\ 0 \end{bmatrix} \\
 &= -\frac{8}{3}.
 \end{aligned}$$

3. *Problem 3 (30 points).* A variant on the task assignment problem.

In the task assignment problem, we have n people and n tasks. Each person must be assigned to a task, and each task must be assigned to only one person. Additionally, for every assignment j of a specific person to a task, we incur cost c_j . The goal of this problem is to match people with tasks while minimizing the sum of costs.

- (a) (15 points) We can formulate this problem as a minimum cost network flow problem by setting up a network with n nodes representing the people, and n other nodes representing the tasks. Every arc i connects a person to a task and has cost c_i . Each person node has a *supply* of 1, and each task node has a *demand* of -1 . The following image shows a simple example with $n = 2$. The people are labeled p_1, p_2 and the tasks are labeled t_1, t_2 . The supply and demands are written on the dashed arrows, and the costs are written *above* the solid arcs.



Formulate this specific example as a minimum cost network flow problem. That is, give c, A, b so that the optimization problem can be written as

$$\begin{aligned} & \text{minimize} && c^T x \\ & \text{subject to} && Ax = b \\ & && 0 \leq x \leq \mathbf{1}. \end{aligned} \tag{1}$$

- (b) (10 points) Now suppose we add a new task t_3 but we maintain that each person can only perform one task, which means all tasks cannot be assigned. The cost associated with the arc connecting p_1 to t_3 is 7 and of the arc connecting p_2 to t_3 is 6. If a task t_j remains unassigned, then we pay a penalty cost d_j . Let $d_1 = 10$, $d_2 = 11$, and $d_3 = 12$.

Our goal is to now minimize the sum of assignment costs plus any incurred penalties. How would you modify the network in part (a) to answer this new question? You may draw a new network for your answer instead of defining new matrices.

Hint: We need a way to encode an unassigned task. One approach is to introduce an imaginary person p_3 and connect it by specifying arc costs and capacities.

- (c) (5 points) What if you have another new task t_4 ? How would the network change? A word description of how you would change the network is sufficient.

Solution.

- (a) There are 4 nodes and 4 arcs, so $A \in \mathbf{R}^{4 \times 4}$. Assuming we order the nodes p_1, p_2, t_1, t_2 , and the arcs $(p_1, t_1), (p_1, t_2), (p_2, t_1), (p_2, t_2)$, then we get the following:

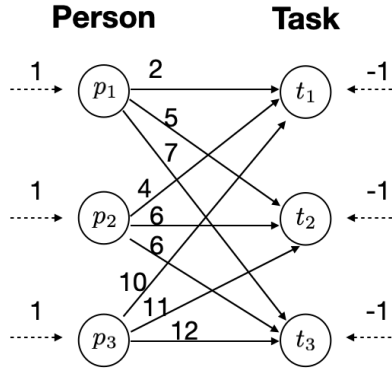
$$c = (2, 5, 4, 6)$$

$$A = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ -1 & 0 & -1 & 0 \\ 0 & -1 & 0 & -1 \end{bmatrix}$$

$$b = (1, 1, -1, -1).$$

- (b) We introduce an imaginary person denoted by p_3 . We will represent a task remaining it unassigned by assigning it to our imaginary person. For this representation, we set the costs for p_3 equal to the penalty if the task is unassigned, i.e., $c(p_3, t_j) = d_j$ for $j = 1, 2, 3$. This way, a task is either assigned to exactly one person or it is unassigned (i.e. assigned to the imaginary person). Setting $c(p_3, t_j) = d_j$ allows us to incur the penalty cost for unassigned tasks.

The new network would look like:



- (c) Similar to part (b), we add a new task t_4 and label the new arcs with cost $c(p_i, t_j)$ for $i = 1, 2$ and $j = 1, 2, 3, 4$. To handle the penalties, we set $c(p_3, t_j) = d_j$ as before. However, in the current formulation, the imaginary person will only get one task, but we need to represent multiple unassigned tasks. There are two equivalent ways to do this:

- Create another imaginary person p_4 and set $c(p_4, t_j) = d_j$. This way, the tasks assigned to either p_3 or p_4 represent unassigned tasks.
- Increase the supply to node p_3 from 1 to 2, which allows us to represent 2 unassigned tasks total.